Key features of quadratic functions

1 Find the value of the quadratic expression $3x^2$ for each of the following x-values.

a $x = 5$

b $x = -5$

2 Find the value of the following quadratic expressions given x.

a $3x^2 + 1, x = 1$

b $4x + 5x^2, x = 7$

c $6x^2 + 5x - 4, x = 8$

d $3x^2 - 10x + 3, x = 3$

3 Determine whether or not the following are quadratic expressions.

a $6x^2 - 2x + 7$

b $-2x + 20$

c $-3x^2 + 4$

d $7x^3 + 3x^2$

e $(9x + 4)^2$

f $\left(x^2 + 7\right)^2$

g $\sqrt{x^2 + 3x - 4}$

h $\dfrac{1}{4x^2 - 6x - 6}$

i $x^2 + \dfrac{1}{x}$

4 Determine whether or not the following equations represent a quadratic relationship between x and y.

a $2x - 3y - 2 = 0$

b $y = 2x^2 - 3x - 2$

c $y = 5x^3 - x^2$

d $y = x^2 - 1$

e $y = 2x - 3$

f $y = \left(x^2 - 3\right)^2$

g $y = 1 - 3x - 5x^2$

h $(x - 3)^2 = y$

i $y = x^2 + 4x$

j $y = \dfrac{5}{x^2}$

k $y = 1 - x^2$

l $y = \left(x^2 - 1\right)^3$

m $y = x^2 + \dfrac{2}{x}$

n $y = \sqrt{x^2 + 2}$

o $y = 2x^2$

5 State whether the following values of x, when substituted into $x^2 + 5x - 36$, give a value of 0.

a -4 and 9

b 4 and -9

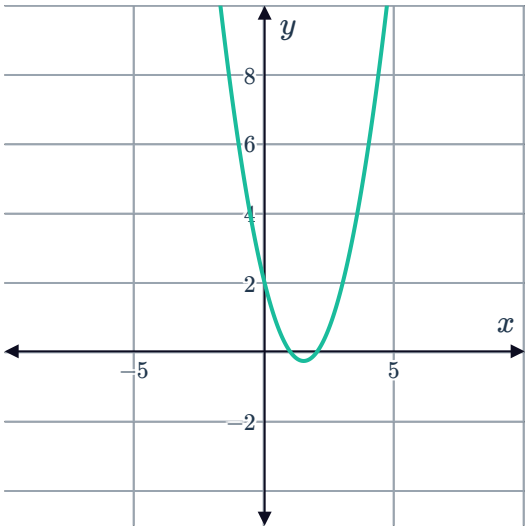
c 8 and 6

d -9 and -4

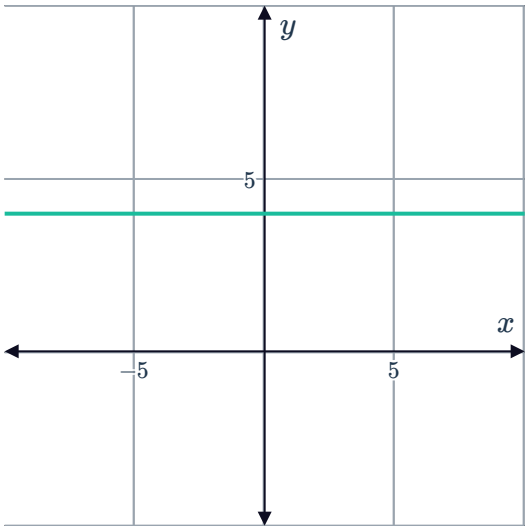
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6 For the following graphs, identify the relationship represented as linear, quadratic or neither.

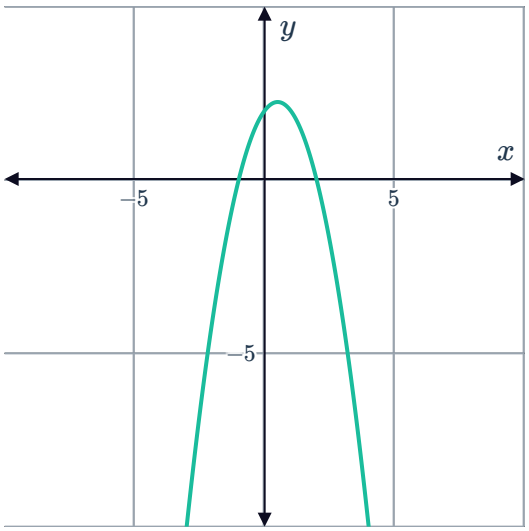
a



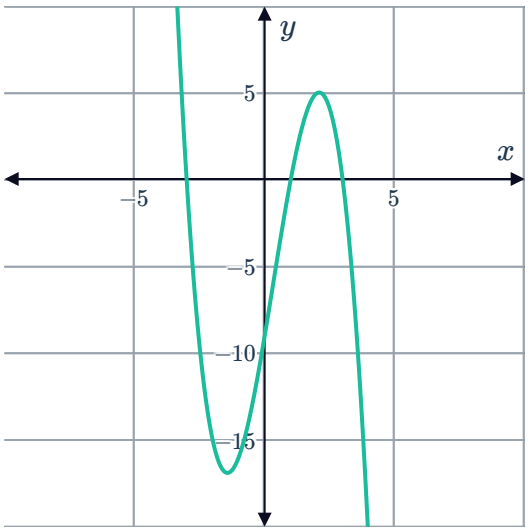
b



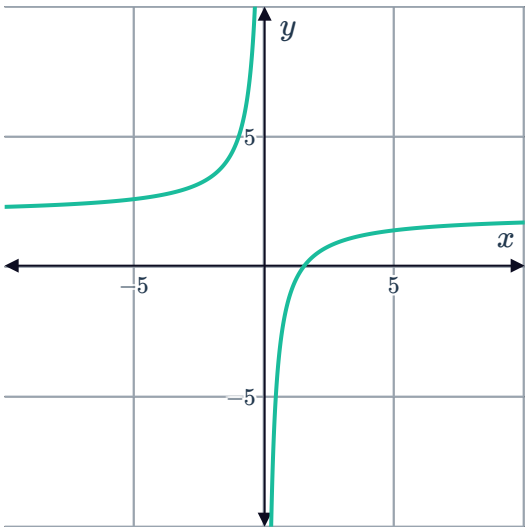
c



d



e



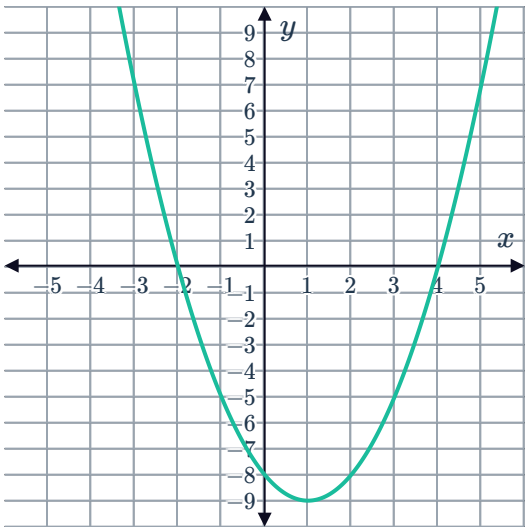
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7 Consider the following graphs.

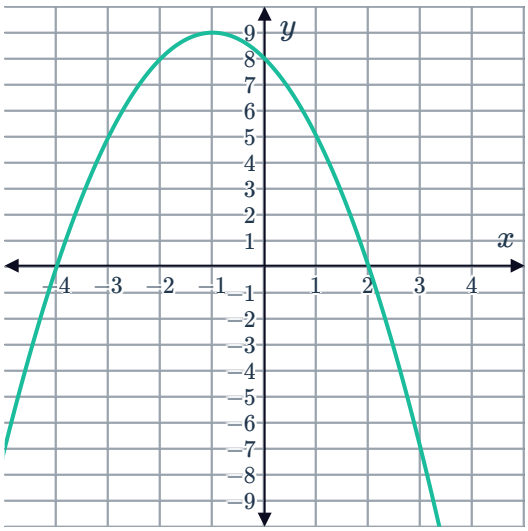
i Find the coordinates of the x-intercept(s) in the form (a, b).

ii Find the zero(s) of the function.

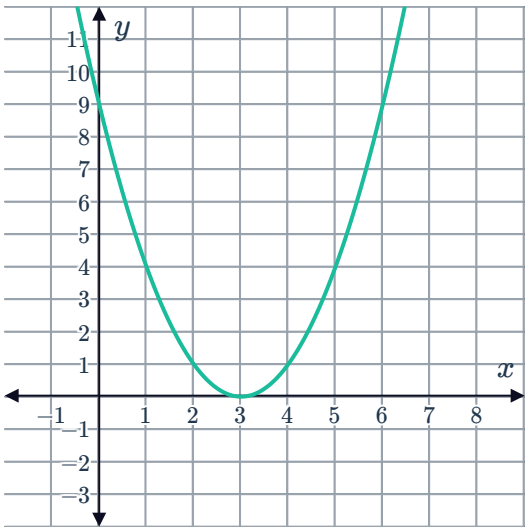
a



b



c



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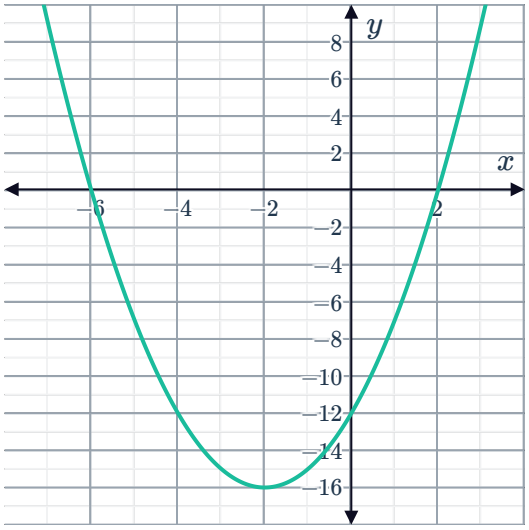
8 Consider the following graphs:

i Find the x values of the x-intercepts of the graph.

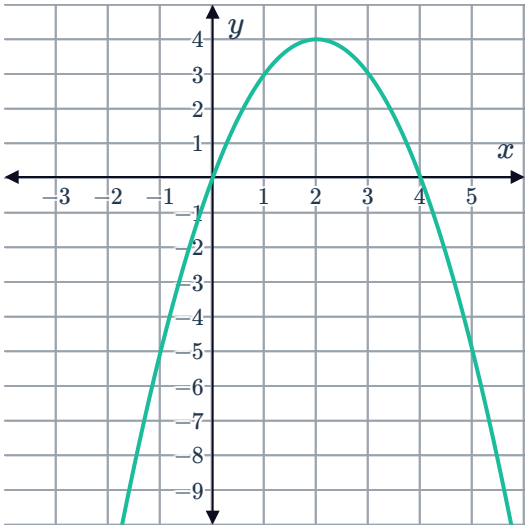
ii Find the y value of the y-intercept of the graph.

iii Does the graph have a maximum or a minimum value? Find the value.

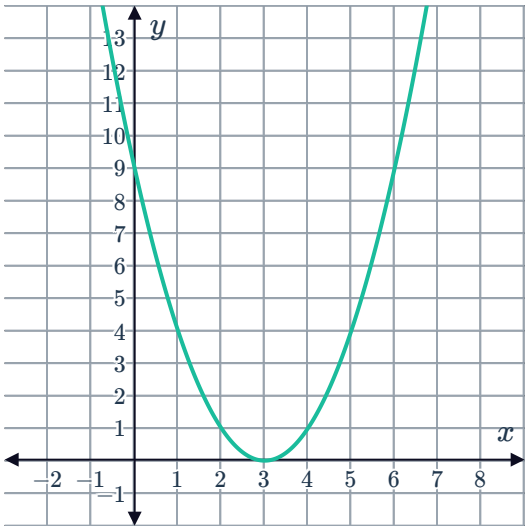
a



b

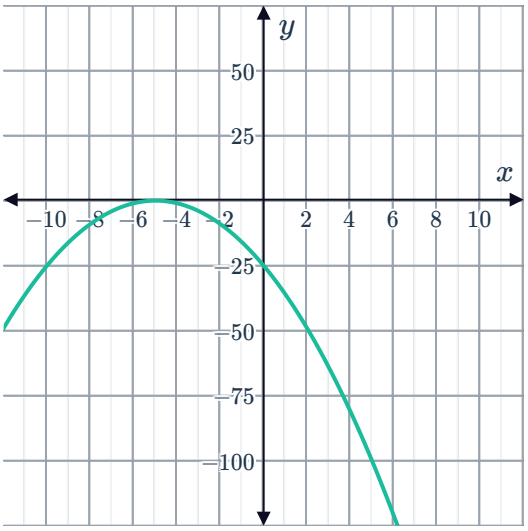


c



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9 Consider the given graph:



a Find the x-value of the x-intercept of the graph.

b Find the y value of the y-intercept of the graph.

c Find the absolute maximum of the graph.

d Determine the interval of x in which the graph is increasing. Give your answer as an inequality.

10 Consider the quadratic function defined in the table.

x	-7	-6	-5	-4	-3	-2	-1
y	11	6	3	2	3	6	11

a Find the coordinates of the vertex in the form (a, b).

b Is the vertex a maximum or minimum point?

c Find the maximum or the minimum value of the function.

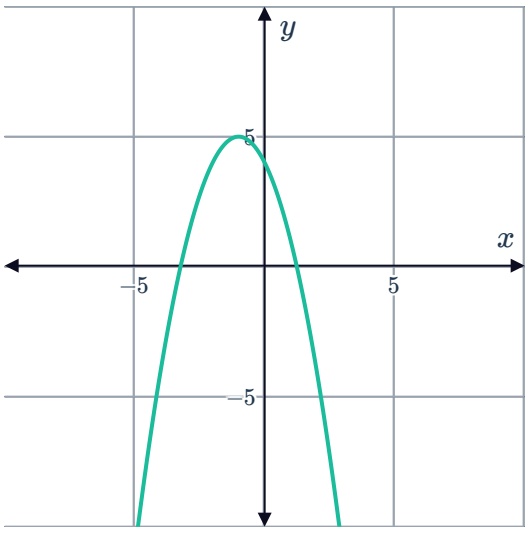
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11 Consider the following parabolas of the form $y = ax^2 + bx + c$, where $a > 0$:

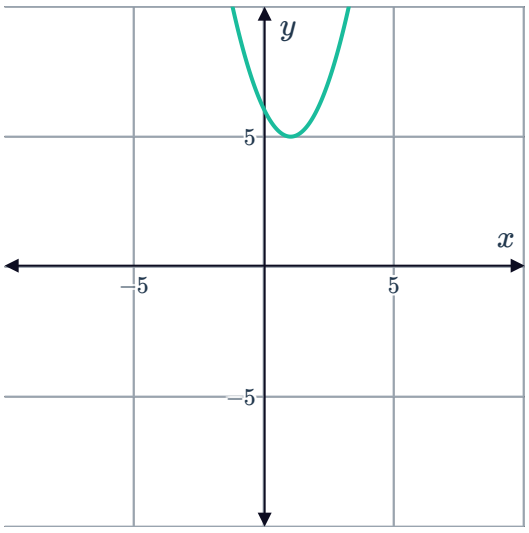
a State whether the vertex is a maximum or minimum point.

b State whether or not the following could be the graph of $y = ax^2 + bx + c$ for which $a > 0$.

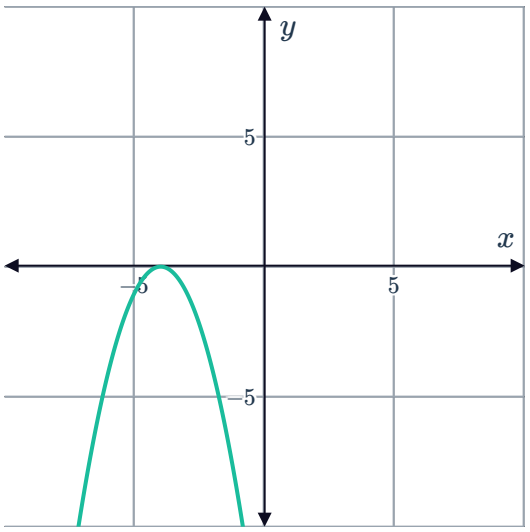
i



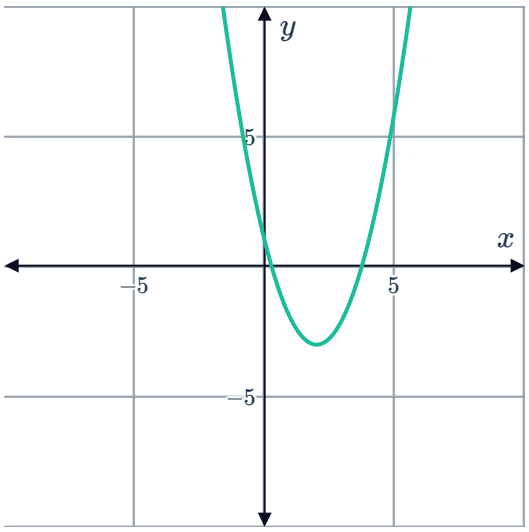
ii



iii



iv



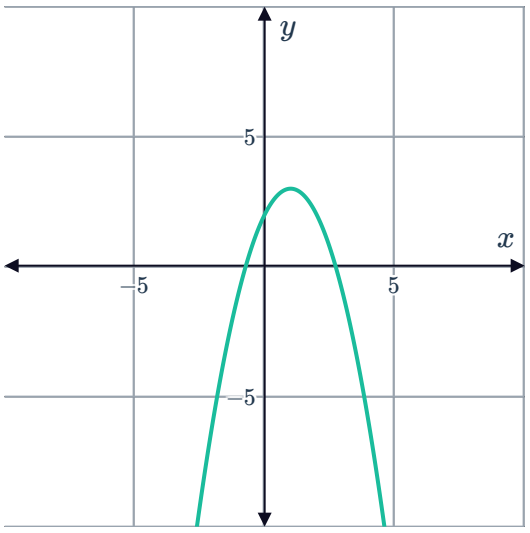
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12 Consider the following parabolas of the form $y = ax^2 + bx + c$, where $a < 0$:

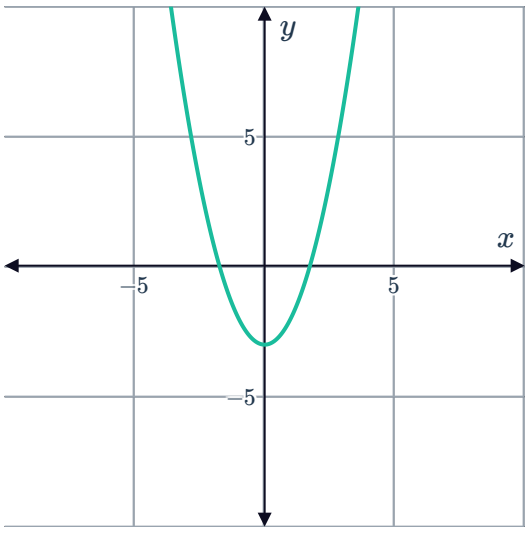
a State whether the vertex is a maximum or minimum point.

b State whether or not the following could be the graph of $y = ax^2 + bx + c$ for which $a < 0$.

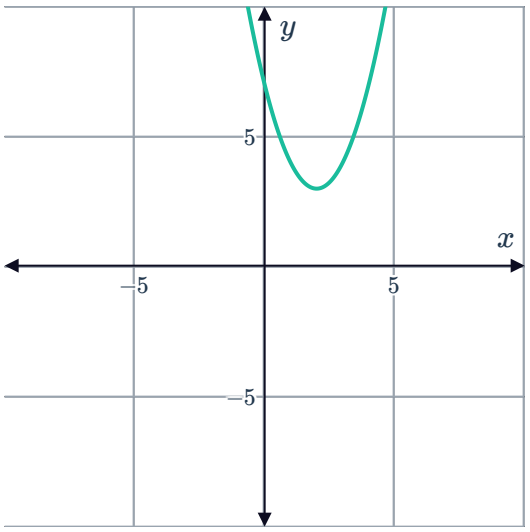
i



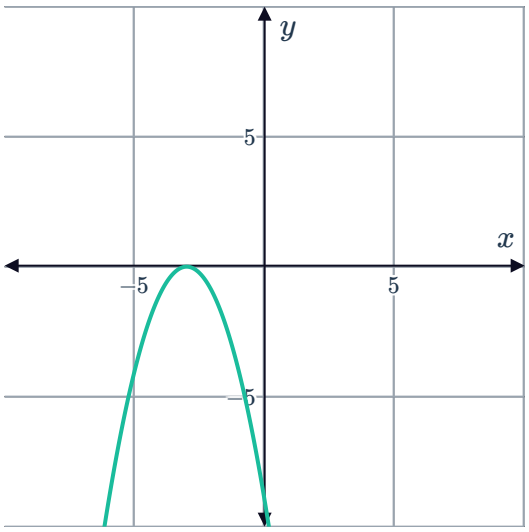
ii



iii

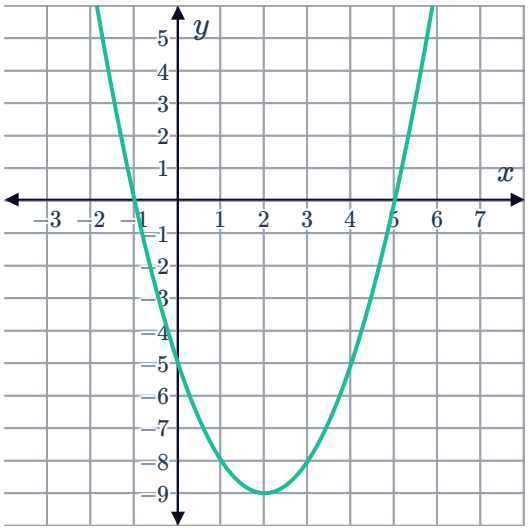


iv



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13 Consider the graph of the quadratic function.
Determine the values of x for which the function is decreasing.



14 Consider the following diagram which shows only one half of the graph of a quadratic function.

a Complete the table of values for the equation.

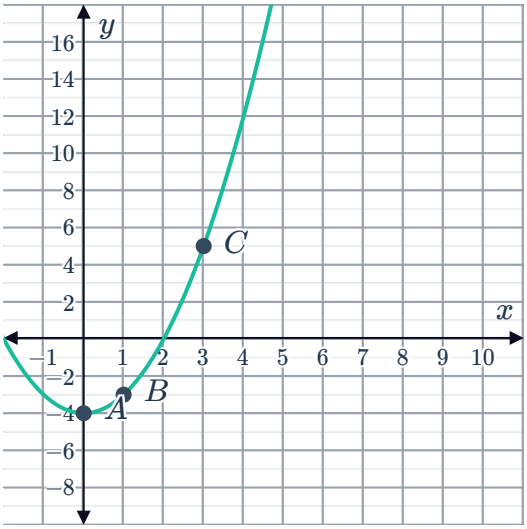
x	-3	-2	-1	0	1	2	3
y							

b Determine the x values of the x-intercepts.

c State the coordinates of the vertex in the form (a, b).

d State whether the vertex represents a minimum or a maximum.

e If we knew the equation of the curve, what would be the sign of the leading coefficient?



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15 Consider a quadratic function defined by the following tables:

i Write the coordinates of the vertex in the form (a, b).

ii State whether the vertex represents a minimum or a maximum.

iii Determine the maximum or the minimum value of the function.

a

x	0	1	2	3	4	5	6
y	-7	-2	1	2	1	-2	-7

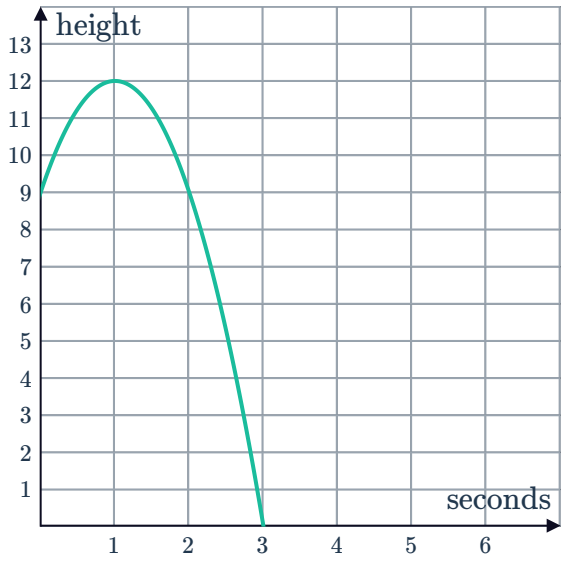
b

x	2.5	3	3.5	4	4.5	5	5.5
y	14	9	6	5	6	9	14

c

x	-5.5	-5	-4.5	-4	-3.5	-3	-2.5
y	-21	-11	-5	-3	-5	-11	-21

16 Consider the given graph which shows the height of a soccer ball after it is kicked.



a What type of function is this?

b Find the y-value of the y-intercept.

c What does the y-intercept represent?

d Find the x-value of the x-intercept.

e What does the x-intercept represent?

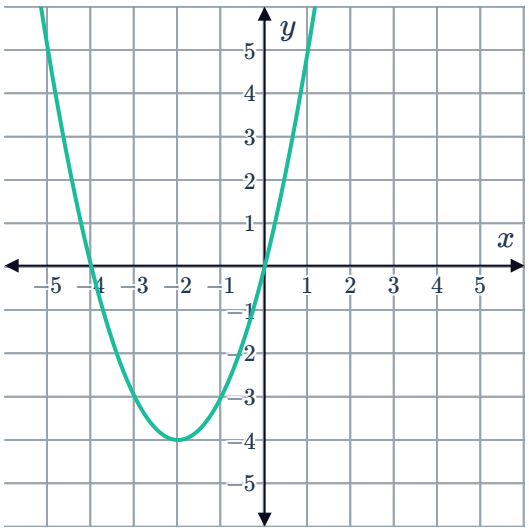
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17 Consider the function of the form $y = f(x)$ defined by the following graphs:

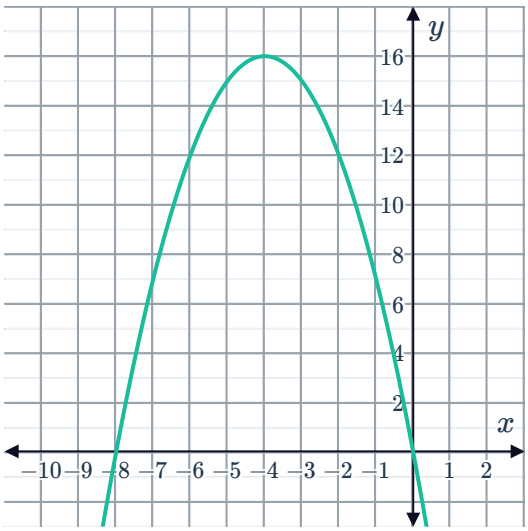
i Does the graph have a maximum or a minimum value? Find the value.

ii Hence determine the domain and range of the function.

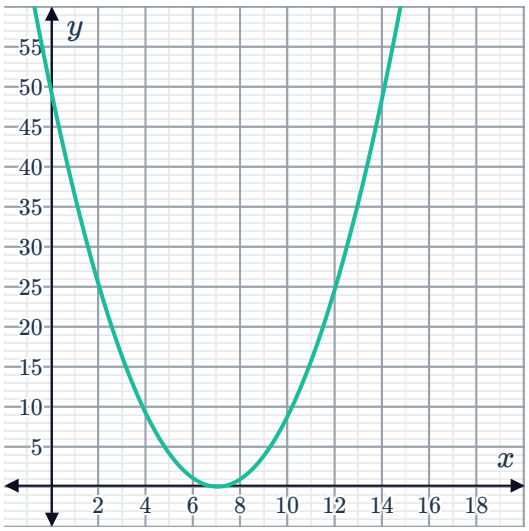
a



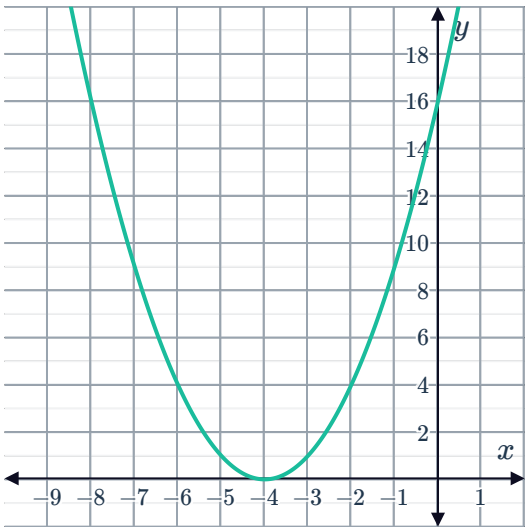
b



c



d



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18 Consider a quadratic function that passes through the points shown in the following tables.

i Does the function have a maximum or minimum value? Find the value.

ii Find the range of the quadratic function as an inequality in terms of y.

a

x	1	2	3	4	5	6	7
y	17	7	1	-1	1	7	17

b

x	-9	-8	-7	-6	-5	-4	-3
y	-11	-6	-3	-2	-3	-6	-11

c

x	5.5	6	6.5	7	7.5	8	8.5
y	15	5	-1	-3	-1	5	15

d

x	-5.5	-5	-4.5	-4	-3.5	-3	-2.5
y	-12	-7	-4	-3	-4	-7	-12

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